

PUSHKAR SINGH

Data Scientist | Machine Learning Engineer

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SUMMARY

Final-year B.Tech student in Computer Science and Data Science who builds end-to-end machine learning systems across computer vision, speech and large language models. Built and deployed three AI applications, including a GPU-served computer vision platform and an on-device voice AI agent with 3,700+ automated tests. Seeking Data Scientist, Machine Learning Engineer and AI Engineer internship and full-time roles.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, C

Machine Learning and Deep Learning: PyTorch, Convolutional Neural Networks (CNN), BiLSTM with Attention, Transformers, Transfer Learning, Data Preprocessing, Data Augmentation, Model Evaluation

Computer Vision: Object Detection (YOLOv8), Vision-Language Models (Florence-2), OCR (PaddleOCR), Image Classification, Document Layout Analysis

NLP, Speech and LLMs: Large Language Models (Gemini API, Qwen3, llama.cpp), Prompt Engineering, Speech Recognition (Whisper), Text-to-Speech, Voice Activity Detection, Speaker Verification

MLOps and Deployment: FastAPI, REST APIs, CUDA, Google Cloud Run (GPU), Vercel, Render, Git, Git LFS, pytest, Linux

Web, Databases and Tools: React, Node.js, Express.js, MongoDB, PostgreSQL, SQLite, Kaggle, GitHub

PROJECTS

VisionAssist AI - Multi-Model Computer Vision Platform | 2026

Python, PyTorch, PaddleOCR, YOLOv8, Florence-2, FastAPI, React, TypeScript, Google Cloud Run

- Built a voice-first vision platform where a trained gatekeeper classifier routes each image to the right specialist pipeline across five image types (document, scene, screenshot, chart, medical).
- Developed a document pipeline with PaddleOCR, layout analysis and a custom BiLSTM with attention document-type classifier, trained on Kaggle GPUs with data preprocessing and augmentation.
- Integrated Florence-2 for scene captioning and YOLOv8 for object detection; a fusion engine merges six models' outputs into one spoken summary.
- Engineered a GPU-memory-aware model loader (lazy loading and eviction under 6 GB VRAM); deployed on Google Cloud Run with an NVIDIA L4 GPU and FP16 inference at under 2 seconds per image.

Harvey - Local-First Personal AI Agent for Windows | 2026

Python, llama.cpp, Qwen3, faster-whisper, Kokoro TTS, Silero VAD, Playwright, SQLite, pytest

- Designed a deterministic-first agent architecture (reasoner, planner, security authority, execution guard) where a local LLM (Qwen3-1.7B on llama.cpp) is a fallback that cannot bypass safety checks.
- Built a fully on-device voice pipeline: wake word, Silero VAD, faster-whisper speech recognition and Kokoro neural text-to-speech.
- Implemented speaker verification with ECAPA voice embeddings so only the owner's voice can issue commands.
- Wrote 63,000+ lines of Python with 3,700+ pytest tests; integrated Playwright automation, Gmail and Calendar APIs, and SQLite memory.

Genesis Prep AI - LLM-Powered Interview Preparation Platform | 2026

Gemini API, React, Node.js, Express.js, MongoDB, JWT | Live: genesis-prep-ai-2.onrender.com | Code: github.com/PushkarSingh0074/genesis-prep-ai

- Used Gemini 2.5 Flash to generate company-specific interview questions across tracks and difficulty levels.
- Designed an LLM evaluation rubric scoring technical depth, communication, confidence and problem solving, with strengths, gaps and a hiring verdict.
- Added voice interviews with live transcription (Web Speech API) and a dashboard tracking scores, streaks and readiness; secured with JWT and bcrypt, deployed on Render with MongoDB Atlas.

EDUCATION

Bachelor of Technology in Computer Science and Data Science | 2023 - 2027 (Expected)

Galgotias College of Engineering and Technology | CGPA: 6.8/10

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks

ADDITIONAL INFORMATION

Preparing for GATE CSE. Interests: Computer Vision, Deep Learning, Applied Machine Learning Systems, Open Source.